

Disproof of Conjecture 00000001046:

The differential uniformity of $x \mapsto x + x^{q-2}$ is not 2 for large q

TLMC falsification pack

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Verdict

FALSE.

1 The conjecture, verbatim, and the object

Conjecture 00000001046. *The differential uniformity of $x \mapsto x + x^{q-2}$ is 2 for large q (uniqueness of the optimal almost perfectly nonlinear (APN) function for large q).*

Object discipline. We attack exactly the named object. Let $F_q = \text{GF}(2^m)$, $q = 2^m$, and

$$f: F_q \rightarrow F_q, \quad f(x) = x + x^{q-2}.$$

For $x \neq 0$, $x^{q-2} = x^{-1}$, and $f(0) = 0$ because $0^{q-2} = 0$ ($q - 2 \geq 1$); thus f is inversion with $0 \mapsto 0$, plus the identity — precisely the map $x \mapsto x + x^{q-2}$ of the conjecture. Its *differential uniformity* is the standard quantity

$$\delta(f) = \max_{a \in F_q^*, b \in F_q} \#\{x \in F_q : f(x+a) + f(x) = b\},$$

the additive subtraction being XOR in characteristic 2. The conjecture asserts that $\delta(f) = 2$ for all sufficiently large q .

2 The attack

Take the input difference $a = 1$ and output difference $b = 0$. For $x, x+1 \neq 0$,

$$f(x+1) = f(x) \iff (x+1) + (x+1)^{-1} = x + x^{-1} \iff 1 = x^{-1} + (x+1)^{-1} = \frac{1}{x(x+1)},$$

i.e. $x^2 + x + 1 = 0$. In $\text{GF}(2^m)$ this quadratic has exactly two roots iff $3 \mid 2^m - 1$, i.e. iff m is even. The two remaining values $x = 0$ and $x = 1$ are degenerate solutions as well ($f(0) = 0 = 1 + 1 = f(1)$, using $0^{q-2} = 0$ and $1^{-1} = 1$).

Fact 1. *For every even m : $\#\{x \in F_{2^m} : f(x+1) + f(x) = 0\} = 4$, hence $\delta(f) \geq 4 \neq 2$ at $q = 2^m$.*

The two non-degenerate solutions are the primitive third roots of unity ω, ω^2 (so the four solutions are $0, 1, \omega, \omega^2$); at $q = 16$ (poly $X^4 + X + 1$) they evaluate to $x \in \{0, 1, 6, 7\}$. The count of a derivative entry is invariant under field isomorphism (an isomorphism preserves $+$, $\cdot$, $^{-1}$, hence conjugates f to itself), so the witness count 4 at $(a, b) = (1, 0)$ does not depend on the chosen irreducible polynomial.

2.1 Full enumeration (machine recomputation)

m	q	$\delta(f)$	parity of m
4	16	4	even
5	32	2	odd
7	128	2	odd
8	256	4	even
9	512	2	odd
10	1024	4	even

Each entry is a complete enumeration over all $a \in F_q^*$, $b \in F_q$, $x \in F_q$ (`reproduce.py`; field polynomials verified irreducible: X^4+X+1 , X^5+X^2+1 , X^7+X+1 , $X^8+X^4+X^3+X+1$, X^9+X^4+1 , $X^{10}+X^3+1$). Even m produces arbitrarily large q with $\delta(f) = 4 \neq 2$; therefore the conjectured threshold does not exist and the conjecture is false.

3 Formal certificate (Lean 4)

`lean4/Main.lean` (checked by `lean4/Check.lean`) certifies the attack on concrete fields, everything table-driven and closed by kernel `decide` on concrete `Nat` arithmetic:

- `TLMC1046.delta16_eq_4`: at $q = 16$, every derivative value occurs at most 4 times over all $a \neq 0$ and $(1, 0)$ occurs exactly 4 times — the complete computation of $\delta(f) = 4$;
- `TLMC1046.cnt256_1_0` and `TLMC1046.cnt1024_1_0`: the 4-fold occurrence of $(1, 0)$ persists at $q = 256$ and $q = 1024$;
- `TLMC1046.disproof` packages the exhibits.

There is no `sorry` and no `native_decide`; the axiom audit over all 9 theorems reports 9/9 axiom-free (no `propext`, `Quot.sound`, `Classical.choice`).

4 Boundaries

For odd m the conjectured value $\delta(f) = 2$ genuinely holds (verified at $q = 32, 128, 512$); the failure is entirely the even- m family (Fact 1), consistent with the classical fact that $x + x^{-1}$ is APN iff m is odd. We claim nothing about APN-uniqueness beyond the refuted premise $\delta(f) = 2$ for large q .